

Generative AI

1. General

1.1 Course Number: CS441

1.2 Contact Hours: 2-0-2 Credits: 8

1.3 Semester-offered: 4th Year (CSE, IT, CSD, IDD)

1.4 Prerequisite: None

1.5 Syllabus Committee Member: Dr. Pallabi Saikia

2. Objective: To introduce the students to the basics of both theoretical and practical aspects of broader area of Electronics Engineering

3. Course Content:

Unit-wise distribution of content and number of lectures

Unit	Topics	Sub-topic	Lectures
1	Mathematical Foundations & Probabilistic Models	Probability and latent variable modeling. Variational Autoencoders (VAEs), KL divergence. Normalizing Flows and autoregressive models. Energy-based models and modeling densities.	5
2	GANs and Score-Based Diffusion	GAN architecture and training dynamics. Advanced GANs: StyleGAN, CycleGAN, BigGAN. Score-based generative models and denoising diffusion. Applications in creative content and synthetic data.	5
3	Transformers and Foundation Models	Attention mechanism and Transformer architecture. Self-supervised learning and pretraining strategies. Foundation models: GPT, BERT, CLIP, DALL·E, Latent Diffusion, Mixture-of-Expert and Mixture-of-Recursions. Multimodal learning and transfer learning.	5
4	Prompting, Fine-Tuning and Retrieval	Prompt engineering and prompt-tuning techniques. Parameter Efficient Fine-Tuning: LoRA, QLoRA, Soft Prompt Tuning, BitFit, Adapters. Retrieval-Augmented Generation (RAG) pipelines. RLHF, reward models, and model alignment. Context Engineering.	7

5	Deployment, Ethics & Real-World Case Studies	LLM deployment, monitoring, cost, and scaling. Bias, fairness, hallucination, and regulatory issues. AI in healthcare, finance, e-commerce, education. Ethical safeguards and model auditing practices.	7
		Total	27

4. Readings

4.1 Textbook:

- i. Sandipan Dey, *Hands-On Generative AI with Transformers and Diffusion Models*, Packt, 2023
- ii. Dhamani, Numa, and Maggie Engler, *Introduction to Generative AI*, Manning Publications, 2023
- iii. David Foster, *Generative Deep Learning: Teaching Machines to Paint, Write, Compose, and Play*, O'Reilly, 2nd Ed., 2022

4.2 Reference books:

- i. Goodfellow, I., Bengio, Y., & Courville, A. (2016). *Deep Learning*. MIT Press.
- ii. Chollet, F. (2021). *Deep Learning with Python*. Manning Publications.
- iii. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). "Attention Is All You Need." *Advances in Neural Information Processing Systems (NeurIPS)*.
- iv. Kingma, D. P., & Welling, M. (2013). "Auto-Encoding Variational Bayes." *International Conference on Learning Representations (ICLR)*
- v. Radford, A., et al. *Generative Pretraining from Pixels (DALL-E paper) and Learning Transferable Visual Models From Natural Language Supervision (CLIP paper)*.
- vi. Brown, T. B., et al. *Language Models are Few-Shot Learners (GPT-3 paper)*.

5. Outcome of the Course:

The student will learn about the fundamentals of Generative AI, including various models, architectures, and their applications. They will gain practical experience in building, fine-tuning, and deploying generative models, as well as understanding ethical considerations and real-world case studies.